AUROVILLE ECO-SERVICE ERP

Open Source Architecture & Development Transfer Proposal

INITIATIVE SPONSOR	Dmitry Lab
TARGET RECIPIENT	Auroville Eco-Service
DOCUMENT STATUS	Revised proposal — Open Source, no-obligation framework
DOCUMENT PURPOSE	Free-of-charge transfer of the current development to Auroville Eco-Service for independent review, use and further adaptation.

THE CORE OFFER

Dmitry Lab makes the current development available to Auroville Eco-Service free of charge. The contribution is intended for independent review, use, adaptation and further development at the organisation’s discretion. No purchase, paid service, implementation commitment or continuing relationship is required.

The materials describe an engineering foundation in progress—not a completed, production-ready ERP and not a system already deployed in Auroville.

1. EXECUTIVE SUMMARY

This proposal accompanies the free-of-charge transfer of the current Dmitry Lab development related to a waste-management and resource-accounting ERP concept for Auroville Eco-Service. The contribution brings together an architectural design, an implemented database structure, selected software components, technical documentation and a synthetic-data utility.

The purpose is practical: to place the existing engineering work at the organisation’s disposal so that its team can inspect the structure, evaluate its relevance, reuse suitable parts or adapt the work independently. The proposal does not assume that Eco-Service will adopt the system or proceed with implementation.

NO PAYMENT • NO PROCUREMENT OBLIGATION • NO IMPLEMENTATION OBLIGATION

The current development is offered without charge. Auroville Eco-Service remains free to use the materials to the extent permitted by the applicable project and third-party licences, modify or extend them, select only useful components, or decide not to proceed.

2. CURRENT DEVELOPMENT: SCOPE AND STATUS

The present contribution is an engineering base with components at different maturity levels. The distinction below is important when evaluating what is available now and what would require additional work.

AREA	CURRENT STATE
Architecture and data model	Designed structure for operational entities, relationships and event-based records.
Database	Implemented database structure / SQL definitions for core entities and operational records.
Backend components	Selected Python components, including asynchronous database access and data-validation models.
Application layer	Selected Reflex-based components and interface specifications; not every described interface feature is a completed production function.
Testing utility	environment_initializer.py for generating synthetic records in a controlled test environment.
Deployment materials	Docker-related materials, README and technical setup guidance; target-environment validation remains necessary.

IMPORTANT STATUS LIMIT

The current work is not a full production ERP installation. No live operational deployment or field validation at Auroville Eco-Service is claimed.

3. OPERATIONAL PROCESS MODEL

The system concept is organised around recording operational events and associating them with identifiable operational objects. The following sequence is an illustrative process model for discussion; it is not a claim that every stage or workflow has already been implemented or confirmed by Eco-Service.



3.1 Event history

Rather than relying only on a single overwritten status, an event-based approach can preserve a sequence of recorded changes. Example event labels in the design include:

EVENT	OPERATIONAL MEANING
SUBMITTED	A collection, load or material record is entered into the system.
ARRIVED	The relevant load or item is recorded as having reached a receiving point.
WEIGHED	A weight measurement is associated with the relevant record and context.
SORTED	A sorting action or resulting material fraction is recorded.

3.2 Material accounting

A structured record can support reconciliation between incoming material, sorted fractions, residuals, transfers and stock changes. A reliable material balance still depends on the quality and completeness of operational measurements.

> **DATA INTEGRITY IS NOT THE SAME AS OPERATIONAL ACCURACY**
>
> The software can enforce structure and validation rules, but it cannot independently guarantee that a recorded weight, classification, moisture level, contamination rate or physical movement is accurate. These depend on instruments, procedures and operator practice.

3.3 Contextual records

The architecture uses relational records alongside event-specific contextual data (including JSONB where appropriate). This allows events to retain relevant details without forcing every event type into an identical fixed set of fields. Such fields should be governed by validation rules and clear documentation.

4. TECHNICAL ARCHITECTURE

LAYER	TECHNOLOGY / APPROACH	ROLE
Database	PostgreSQL; PostGIS where spatial data is required	Relational data foundation and optional spatial capability.
Data access	Python; asyncpg	Asynchronous application-to-database access.
Validation	Pydantic	Structured validation of application data.
Application UI	Reflex components	Selected administrative interface components.
Field / mobile direction	PWA specifications	Browser-based access concept for field workflows; device/browser compatibility requires validation.
Messaging interface	Messaging-platform concept	Optional interaction channel; scope and implementation status depend on the relevant components.
Deployment	Docker-related materials	Reproducible environment setup, subject to target-system compatibility.

5. SOFTWARE TEST DATA — NOT FIELD DATA

The project includes environment_initializer.py, a development utility intended to generate synthetic operational records. It can populate a controlled environment for examining database relationships, event logging, validation and application behaviour.

SIMULATION PARAMETER	VALUE DESCRIBED IN PROJECT MATERIALS
Illustrative time window	1 September 2026, 07:00 UTC – 31 October 2026, 21:00 UTC
Generated records	Approximately 580–760
Synthetic material quantity	Approximately 4.8–6.4 metric tonnes

SYNTHETIC DATA ONLY

These figures are generated test data. They are not measurements, forecasts or operational statistics from Auroville Eco-Service and must not be used as a baseline for procurement, staffing or financial planning.

Software simulation and a field pilot are two different activities. Simulation checks software behaviour under controlled conditions. A field pilot would involve actual operations at an approved site and would require a separate decision by Auroville Eco-Service. No field pilot is represented as having taken place.

6. DEPLOYMENT CONSIDERATIONS

Before any operational use, the target environment would need to be assessed. The architecture alone does not establish that existing infrastructure, devices or connectivity are sufficient.

ITEM	ASSESSMENT REQUIRED
Hosting / server	Determine whether existing organisational infrastructure can host the database and application securely; a new VPS is not assumed to be mandatory.
Domain / access	Use existing services where suitable; define secure access and administrative permissions.
Backups / recovery	Set backup frequency, retention, restore testing and responsibility before critical operational use.
Devices / connectivity	Check available computers, mobile devices, network coverage and browser compatibility at relevant sites.
Identification	Choose site, container or item identifiers only after the operational process is understood.
Scales / instruments	Confirm measurement devices, data capture method, calibration and acceptable tolerances.

7. PROGRESSIVE AUTOMATION AND COST PROFILE

Automation can be introduced incrementally. The levels below describe relative cost and capability, not a mandatory procurement plan or a quotation.

LEVEL	CONFIGURATION	RELATIVE COST PROFILE
1	Manual data entry using available devices	Minimal additional CAPEX if suitable infrastructure exists.
2	Stationary identification: pre-printed placards or fixed codes	Very low additional CAPEX.
3	Reusable QR identification for containers or collection points	Low to moderate CAPEX, depending on devices and labels.
4	Individual bag tracking	Moderate CAPEX plus recurring consumables and handling effort.
5	IoT / automated capture, connected scales or sensors	Higher CAPEX and OPEX, including integration and maintenance.

A portable printer is not inherently required for every QR workflow: fixed identifiers can be printed in advance. The appropriate configuration should follow the process design, not precede it.

A reliable budget requires confirmed process volumes, equipment specifications, staffing assumptions, hosting and connectivity requirements, maintenance expectations and local procurement prices. Total cost of ownership should include recurring costs as well as initial purchases.

8. DATA GOVERNANCE AND SOFTWARE LICENSING

Operational data generated by Auroville Eco-Service should remain under the organisation's governance. Eco-Service should determine its own rules for access, correction, export, retention, backups and administrative control.

- Define user roles and permissions according to operational responsibilities.
- Protect credentials and restrict privileged access.
- Establish backup, recovery and data-export procedures.
- Document how operational records may be corrected while preserving an auditable history.

LICENCE TRANSPARENCY

The project uses Open Source tools and libraries subject to their respective licence terms. The licence applicable to Dmitry Lab's own code must be explicitly identified in the repository. The use of Open Source dependencies, or public visibility of source files, does not by itself determine the licence of the project's own code.

9. TERMS OF THE CONTRIBUTION

FREE-OF-CHARGE TRANSFER OF THE CURRENT DEVELOPMENT

Dmitry Lab is providing the current development materials to Auroville Eco-Service without charging a fee for the contribution. The intention is to make the engineering work available for the organisation's independent consideration and possible use.

Auroville Eco-Service may, at its own discretion and subject to the applicable licences:

- review the architecture, database structure, code and technical documentation;
- use or adapt relevant components for its own purposes;
- continue development independently or with other parties;
- select only those elements that are useful to its operations; or
- decide not to implement the materials.

There is no obligation to purchase services, procure specified equipment, implement the system, accept a prescribed technical configuration or maintain an ongoing relationship with Dmitry Lab.

Questions about the architecture, code, database design, simulation utility and technical assumptions are welcome. If Eco-Service later decides that further technical involvement would be useful, the scope and format of any such discussion can be considered separately, based on actual needs. No further engagement is implied by this transfer.

10. CONCLUSION

This document presents a current engineering development being made available to Auroville Eco-Service free of charge, for independent review and possible use. It is a technical contribution—not a sales offer and not a claim of a completed operational ERP.

THE DECISION REMAINS WITH AUROVILLE ECO-SERVICE

The organisation retains full discretion over whether to use the materials, how to adapt them, whether to continue development and whether to take any further steps.

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